Condition colours

Setting the colour of each condition once, for every figure the session draws.

AT A GLANCE

Where	Data → Colors
Data levels	Protein, phosphosite, phosphoprotein — each with its own palette
Needs	An annotation. The conditions listed here are the conditions it defines.
Applies to	Every plot that groups by condition, at that level
Overridden by	The per-plot condition colours in the plot appearance panel
Persistence	Held for the session, saved as each colour is changed, written into a save state and restored with it.

1 Why a palette rather than per-plot colours

A figure set is read as a set. If the treated condition is orange on the coverage plot and blue on the PCA, every reader of the figure has to re-learn the key on each panel, and at least one of them will misread a result. Setting the colour here costs less than that: one colour per condition, used by every plot at that level for the rest of the session, including the ones drawn before the colour was chosen — they are redrawn when the palette changes.

Each dataset level keeps its own palette, because protein and phosphosite levels may have different conditions. Where they have the same number of conditions, **Copy Colors** carries one palette over to the other level rather than making them match by hand.

2 Setting the palette

1. Choose **Dataset level**. The conditions of the annotation at that level are listed, one row each, in the order the annotation gives.
2. Set a colour either from the swatch, which opens the operating system's colour picker, or by typing a hexadecimal value such as #1F77B4 into the field beside it.
3. There is no save button. Each colour is written as soon as it is complete, and the plots follow immediately.
4. **Reset Default Palette** puts every condition back on the built-in cycle and saves that.
5. With more than one level loaded, choose a level under **Copy colors to dataset level** and press **Copy Colors** to give it the same colours.

Controls on this page

Control	What it does	Default	Where
Dataset level	Which level's palette is being edited. Only the levels actually uploaded are offered.	Protein	page
Colour swatch	The condition's colour, from the system colour picker.	the built-in cycle	page
Hex field	The same colour as #RRGGBB. Anything that is not a complete six-digit value is held while it is being typed and reverts when the field is left.	the built-in cycle	page
Reset Default Palette	Writes the built-in cycle back for every condition at this level, and saves it.	—	page
Copy colors to dataset level	The level **Copy Colors** will write to. Shown only when more than one level is loaded.	the first other level	page
Copy Colors	Copies this level's colours to that level by position: first condition to first condition, and so on.	—	page

3 How a colour is chosen

RESOLUTION ORDER

When a plot needs a colour for a condition it asks three sources in turn, and takes the first answer:

- the condition colours set for *that plot* in the appearance panel, which are part of that plot's own settings;
- this page's palette for that level, if it holds an entry for that condition name;
- otherwise a generated colour, by the condition's position in the list.

The generated colours are a fixed sequence of twenty, beginning #1F77B4, #FF7F0E, #2CA02C, #D62728; beyond twenty conditions they continue by rotating the hue in steps of the golden ratio, alternating saturation and brightness so that neighbours stay distinguishable.

STORAGE

A palette is a map from condition name to a #RRGGBB string, one map per dataset level, held for the browser session. Values are normalised to upper case, and anything that is not six hexadecimal digits is discarded rather than stored. Writing a palette invalidates the cached figures at that level, which is what makes the plots redraw. The palettes of all three levels are written into a save state and come back with it. Reloading the application in the browser starts a new session and clears them, as it clears the uploaded data.

EASY TO GET WRONG

The list on this page shows a repeating cycle of ten colours, while a plot that has not been given a saved palette continues to twenty distinct ones. With eleven or more conditions the swatch shown here for condition eleven onwards is therefore not the colour the plot is using — until something is saved. Pressing **Reset Default Palette**, or setting any one colour, writes the whole list and the plots follow it from then on.

Edge cases

- The palette is keyed by condition name. Renaming a condition in the annotation leaves its old entry behind and the renamed condition falls back to a generated colour, so set it again after a rename.
- **Copy Colors** works by position and refuses when the two levels have different numbers of conditions; it says so rather than copying part of the palette.
- Renaming a condition in the appearance panel changes only a label on one figure. It does not change what this page is keyed on.
- With no annotation there are no conditions and the list says so. Generate the annotation first.

> **WORTH KNOWING**
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> A colour that separates well on screen may not survive print or a colour-blind reader. The safest check is to look at the figure in greyscale — if two conditions become the same grey, they differ only in hue — and to keep the pairs red/green and blue/purple apart in lightness as well as hue. The built-in cycle is designed for this, which is a reason to change as few of its entries as the figure needs.

4 Exporting

There is nothing to download here: the palette is not a file, it travels in the save state and in the figures that use it. To hand the colours to somebody else, save a state; to reuse them at another level, use **Copy Colors**.

Behind this page

Function	What it produces
get_condition_palette	The palette stored for one level
set_condition_palette	Validates and stores it, and invalidates the cached figures
build_condition_color_map	The colour every plot actually uses: stored entry, else generated by position
reset_condition_palettes	Clears the palettes of a session